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Predicting Academic Performance: Machine Learning Insights into GPA Determinants

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***Abstract*—This study looks at predicting student GPA using machine learning models based on academic and non-academic data. Random Forest (RF), Gradient Boosting (GB), AdaBoost, Linear Regression (LR), Ridge, Lasso, Support Vector Regression (SVR), K-Nearest Neighbors (KNN), and XGBoost are among the models evaluated. R-squared (R^2), Mean Squared Error (MSE) and Mean Absolute Error (MAE) are used to assess performance. The most accurate models were Ridge and Linear Regression with accuracy of 95.68%, which suggested a significant linear connection between variables and GPA. Support Vector Regression (SVR) and XGBoost also performed well but did not surpass the linear models. Because of its simplicity and accuracy, the research identifies Ridge Regression as the best model for predicting GPA, providing insightful information that may help educational institutions support student interventions.**

***Index Terms*—GPA Prediction, Machine Learning, Ridge Regression, Linear Regression, Gradient Boosting, Random Forest, Support Vector Regression, XGBoost, Lasso, AdaBoost, K-Nearest Neighbours, Educational Data, Student Performance.**

I. Introduction

Computers do not learn as well as people do, but many machine-learning algorithms have been found that are effective for some types of learning tasks. They are especially useful in poorly understood domains where humans might not have the knowledge needed to develop effective knowledge-engineering algorithms [1], [2]. Generally, machine learning (ML) explores algorithms that reason from externally supplied instances (input set) to produce general hypotheses, which will make predictions about future instances [3], [4].

It is well known that students have some indicators of performance, for example the Grade Point Average (GPA) which is useful for assessing performance in recent educational systems. GPA is one of the key performance indicators of students that are very critical in making decisions on further education, seeking internships, employment, scholarships and other benefits. However, even though there are various identified ways of enhancing academic performance; these include academic input methods such as past grades and non-academic factors such as socio-economic status, attendance, and studying methods, predicting GPA has been a big challenge.

The economic success of any country highly depends on making higher education more affordable and that considers one of the main concerns for any government [5]. Forecasting student performance is essential for educators to obtain early feedback and take immediate action or early precautions if necessary to improve the student's performance. This prediction can be managed by locating the source of the problem. Should it be from extra activities that the student is participating in, family problems, or health problems. All these factors can have a major effect on student performance [6].

To analyze the students' GPA prediction problem, this study addresses the effectiveness of various machine learning models namely, random Forest (RF), Gradient Boosting (GB), AdaBoost, Linear Regression (LR), Ridge, Lasso, Support Vector Regression (SVR), K-Nearest Neighbors (KNN), and XGBoost. Each of these models has its own merits for its capability of handling various data types and variable combinations. The objective of the study is to evaluate such models with a view of employment in determining the students' performance matrices while providing valuable information to the concerned institutions and their teachers.

This research has more significance that goes well beyond enhancing GPA prediction accuracy. It provides a model in which educational institutions can develop their capacity to provide students' support systems. Finding students who are having trouble early and giving them help, like extra lessons, tutoring, and scholarships, can improve their grades and overall performance. This research therefore contributes to the knowledge and practice of the ways of managing education systems and students' success strategies.

II. Literature Review

Machine learning techniques have gained significant attention in educational settings, particularly for predicting student performance. Several studies have explored and compared different algorithms to determine their effectiveness in forecasting student outcomes and supporting timely interventions.

H. Al-Shehri et al. [6] conducted a comparative analysis of Support Vector Machine (SVM) and K-Nearest Neighbor (KNN) models, using a dataset from Portuguese secondary schools to predict student grades. Their findings revealed that SVM slightly outperformed KNN, with correlation coefficients of 0.96 and 0.95, respectively, demonstrating the high predictive power of both models in academic contexts.

Similarly, H.M.R. Hasan et al. [7] utilized machine learning models such as Decision Tree and KNN and predict student performance with an impressive accuracy of 94.44% (Decision Tree) analyzing academic data from 1,170 students, their study focused on identifying at-risk students for early intervention. They proposed that future research could incorporate psychological data to enhance prediction accuracy further.

Building on the theme of early prediction, N. Alangari and R. Alturki [8] explored 15 different machine learning algorithms to predict students' final GPA. Naive Bayes and Hoeffding Tree emerged as the best-performing models, achieving 91% accuracy. The study highlighted the critical role of early course performance in shaping final GPA, offering valuable insights into how early academic indicators can guide student interventions and support.

In contrast, A.S. Hashim et al. [9] found that Logistic Regression was the most accurate model for predicting student performance, with an accuracy of 68.7% for predicting final grades and 88.8% for pass/fail status. This research underscores the potential of machine learning to support educational institutions by flagging students who may require additional assistance to improve their academic outcomes.

Finally, B. Sekeroglu et al. [10] investigated the application of machine learning models such as Support Vector Regression (SVR), Long Short-Term Memory (LSTM), and Backpropagation (BP) to predict and classify student performance. SVR was the best predictor, while BP achieved the highest classification accuracy at 87.78%. This study highlights the growing importance of machine learning in analyzing educational data for more effective student performance predictions.

Additionally, H. Altabrawee et al. [5] highlights the importance of identifying struggling students early for timely support using four machine learning models: ANN, Naive Bayes, Decision Tree, and Logistic Regression. ANN achieved the best accuracy (77%) and ROC index (0.807) and D. Canagareddy et al. [11] uses classification and prediction algorithms to forecast whether students will pass or fail, and estimate their GPA. Algorithms like Naive Bayes, J48, and Random Forest are evaluated, with J48 and Random Forest found to be the most accurate. The model helps universities intervene early for students at risk of failure by analyzing factors such as attendance and grades.

Collectively, these studies demonstrate that various machine learning models, including SVM, Naive Bayes, Logistic Regression, and SVR, are effective in predicting student performance, with the potential to improve academic outcomes through early identification of at-risk students. A comparative analysis of the related works is shown in Table I.

TABLE I
Comparative Analysis of Literature Review

Author	Dataset Size	Machine Learning Algorithms	Best Algorithm
H. Al-Shehri et al., 2017 [6]	395	SVM, KNN	SVM
H. M. R. Hasan et al., 2019 [7]	1170	KNN, DT, RF, GB, SVC	DT
ROMJIST, 2020 [8]	530	15 Classification Algorithms	NB and Hoeffding Tree
A. S. Hashim et al., 2020 [9]	349	DT, NV, LR, SVM, KNN, SMO, NN	LR
B. Sekeroglu et al., 2019 [10]	499	Neural Networks (Back Propagation), SVR, LSTM, SVM, GB	SVR
H. Altabrawee et al., 2019 [5]	161	ANN, NB, DT, LR	ANN
D. Canagareddy et al., 2018 [11]	2000	NB, Logistic Classifier, J48 Classifier, SVM, RF, LR	J48 classifier
S. Kotsiantis et al., 2004 [3]	354 and 28	DT, Neural Networks (Back Propagation), NB, Instance-Based Learning, LR, SMO	NB

III. Methodology

This analysis aims to build machine learning models to predict student GPA using various regression techniques, including ensemble methods and linear models. Figure 1 presents the steps employed in this study.

A. Data Collection and Preparation

The dataset is imported into a pandas DataFrame, where it is cleaned by removing irrelevant columns, such as "StudentID". Categorical variables like "Gender" and "Ethnicity" are converted to numeric using Label Encoding. Numerical features like "Age" and "StudyTimeWeekly" are standardized with StandardScaler to ensure all variables are on the same scale, essential for models like SVR and KNN.

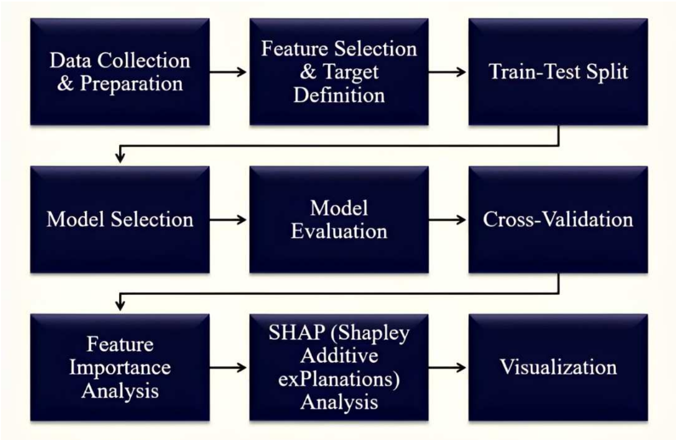

Fig. 1. Overview of the methodology employed in the proposed study

B. Feature Selection and Target Definition

The dataset is divided into features (X) and the target variable (y), with GPA as the target. Features include ”Absences”, ”StudyTimeWeekly”, and ”GradeClass”, excluding GPA from the features to allow the model to learn from other variables.

C. Train-Test Split

The dataset is split into training (80%) and testing (20%) sets to train the model on one portion and evaluate it on another, ensuring generalization to unseen data.

D. Model Selection

A diverse range of models is selected, including Random Forest, Gradient Boosting, AdaBoost, XGBoost, Linear Regression, Ridge, Lasso, SVR, and KNN. Hyperparameter tuning is performed using GridSearchCV combined with 5-fold cross-validation to optimize performance and reduce overfitting.

E. Model Evaluation

Models are evaluated based on R^2 score, Mean Squared Error (MSE) and Mean Absolute Error (MAE). The model with the highest R^2 and lowest MSE is deemed the best for predicting GPA.

F. Cross-Validation

The best model undergoes 5-fold cross-validation to ensure its reliability and generalizability across different data splits.

G. Feature Importance Analysis

Feature importance analysis is conducted for models like Random Forest, XGBoost, and Ridge to identify influential variables, helping interpret model predictions.

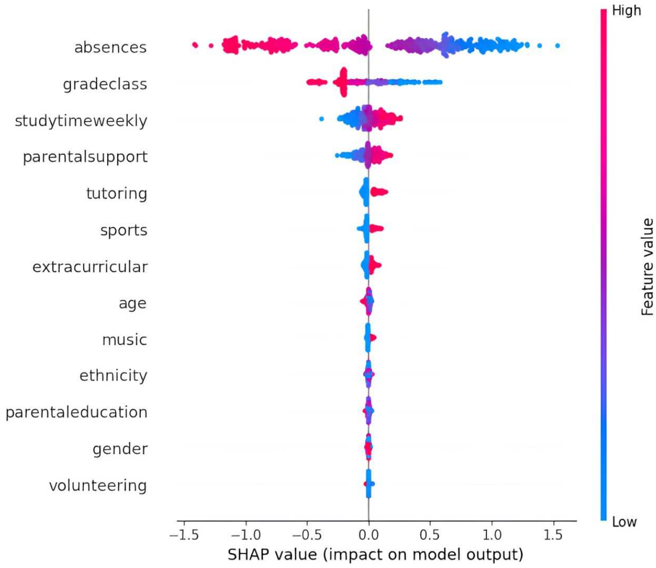

Fig. 2. SHAP values Analysis

H. SHAP (SHapley Additive exPlanations) Analysis

SHAP values are used to visualize the impact of individual features on predictions, indicating which variables, such as ”Absences” and ”StudyTimeWeekly”, significantly influence GPA predictions.

The SHAP summary plot shown in Figure 2 visualizes the impact of individual features on predictions made by a machine learning model. The x-axis represents SHAP values, which indicate the contribution of each feature to the prediction. Positive SHAP values signify that a feature increases the predicted output, while negative values indicate a decrease. The y-axis lists the features in descending order of importance, with the most influential feature placed at the top.

In this plot, *Absences* emerges as the most significant feature, exhibiting a wide range of SHAP values. High absences (in red) generally lead to an increase in the predicted outcome, while low absences (in blue) tend to reduce it. Other notable features, such as *GradeClass* and *StudyTimeWeekly*, also show substantial influence on the model’s predictions. Specifically, *StudyTimeWeekly* demonstrates a mostly positive effect as its value increases.

Features like *Volunteering*, *Gender*, and *Ethnicity* exhibit SHAP values close to zero, suggesting minimal contribution to the model’s predictions. This plot provides a clear representation of how different features, both numerical and categorical, affect the model’s performance.

I. Visualization

The scatter plots for actual vs. predicted GPA. These visualizations aid in understanding model performance and feature significance. Here, we can see the actual GPA vs predicted GPA in scatter plot for every predicted model which we used.

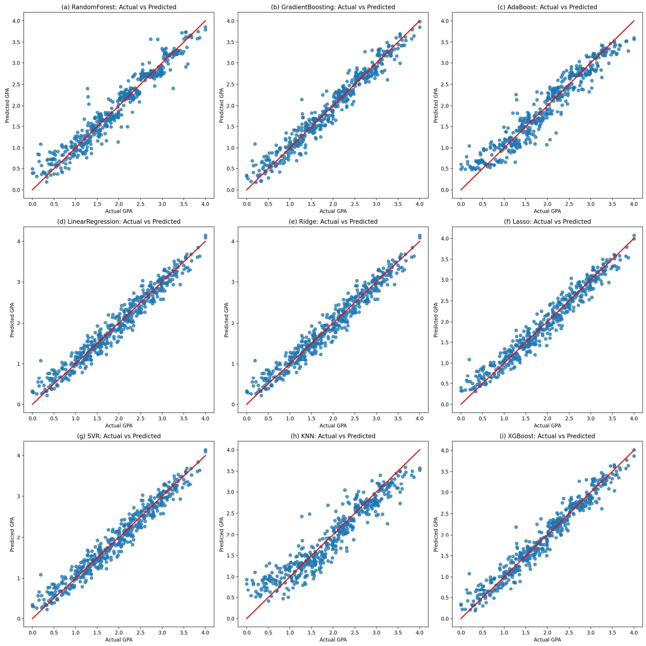

Fig. 3. Scatter Plots for Different Models

The scatter plots shown in Figure 3 display the predicted versus actual GPA for various models, including Lasso, SVR, KNN, XGBoost, Random Forest, Gradient Boosting, AdaBoost, Linear Regression, and Ridge. The red line represents the ideal prediction line where predicted GPA matches actual GPA. All models show close alignment between predicted and actual values, suggesting that both linear and non-linear models capture the same underlying patterns in the data. The similarity in performance across these models indicates that the relationship between features and GPA is highly linear.

IV. Results

The table and graphs in this section showcase a comparative analysis of various machine learning models based on their $\square^2$ values, Mean Squared Error (MSE), and Mean Absolute Error (MAE). Ridge and Linear Regression emerged as the top performers in terms of accuracy, achieving the lowest MSE and MAE. In contrast, K-Nearest Neighbors (KNN) exhibited the lowest performance, with the highest MSE and MAE values.

A. Model Comparison

B. Results Analysis

The result table provided compares the performance of different machine learning models in predicting student GPA using three key metrics: R^2, Mean Squared Error (MSE), and Mean Absolute Error (MAE). Each of these metrics provides insight into the accuracy and reliability of the models.

1. $\square^2$ Value: This metric shows how well the model explains the variance in GPA. An R^2 value closer to 100% means

TABLE II
Model Comparison Based on $\square^2$, MSE, and MAE

Model	**$\square^2$ Value**	**MSE**	**MAE**
Ridge	95.683363%	0.035696	0.150637
Linear Regression	95.683290%	0.035696	0.150639
SVR	95.659228%	0.035895	0.151026
XGBoost	95.576631%	0.036578	0.145224
Gradient Boosting	95.493605%	0.037265	0.146663
Lasso	95.467509%	0.037481	0.151654
Random Forest	93.785145%	0.051393	0.167903
AdaBoost	92.637405%	0.060884	0.192798
K-Nearest Neighbors	89.015029%	0.090838	0.233688

the model is better at predicting GPA. The table shows that Ridge Regression has the highest R^2 value at 95.68%, followed closely by Linear Regression and Support Vector Regression (SVR), which both have R^2 values slightly above 95.65%. These three models demonstrate very strong predictive capabilities, indicating that they capture nearly all the variance in the GPA data. XGBoost, Gradient Boosting, and Lasso also show strong performance, with R^2 values around 95.5%, while Random Forest lags slightly behind at 93.75%. The AdaBoost and K-Nearest Neighbors (KNN) models are less accurate, with lower R^2 values of 92.63% and 89.01%, respectively.

2. MSE (Mean Squared Error): MSE quantifies the average squared differences between predicted and actual values. A lower MSE indicates better performance, as it signifies that the model's predictions are closer to the actual GPA values. Ridge Regression and Linear Regression have the lowest MSE of 0.035696, which correlates with their high $\square^2$ values, showing that these models make more accurate predictions. SVR, XGBoost, Gradient Boosting, and Lasso all show slightly higher MSE values, but remain competitive. The Random Forest model has a noticeably higher MSE of 0.051393, while AdaBoost and KNN models, which are the least accurate in this table, exhibit much higher MSE values of 0.060884 and 0.090838, respectively.

3. MAE (Mean Absolute Error): MAE measures the average magnitude of the errors in the predictions, providing an easy-to-interpret metric of how far off the model's predictions are, on average. In this case, Ridge Regression, Linear Regression, and SVR exhibit the lowest MAE values around 0.1506, suggesting that these models, on average, make small prediction errors. XGBoost, Gradient Boosting, and Lasso show slightly higher but still acceptable MAE values. Random Forest has a higher MAE at 0.1679, while AdaBoost and KNN again display the highest MAE values, indicating larger prediction errors.

C. Analysis and Findings:

Based on the results in the table, Ridge Regression, Linear Regression, and SVR are the top-performing models across all metrics, as they have the highest $\square^2$ values, and the lowest MSE and MAE scores. These findings suggest that simpler linear models with regularization (like Ridge) are highly effective at capturing the relationship between the features and GPA. This performance is consistent with prior research, which often shows that regularized linear models tend to perform well when the underlying relationship between the features and the target variable (GPA) is approximately linear.

For instance, previous studies in educational data mining often find that models like Linear Regression and Ridge perform well in academic performance prediction, especially when the dataset contains both continuous and categorical variables that influence outcomes like GPA. Regularization techniques like Ridge help prevent overfitting, particularly when there are many predictors, as is common in student data. Support Vector Regression (SVR) has also been highlighted in other works for its robustness in handling high-dimensional data and its flexibility in capturing both linear and non-linear patterns. However, despite its strengths, it is computationally more expensive compared to linear models like Ridge and Linear Regression.

XGBoost and Gradient Boosting are powerful ensemble methods that generally outperform linear models in cases where non-linear relationships are dominant. However, in this analysis, both XGBoost and Gradient Boosting have slightly lower R^2 values and higher MSE compared to Ridge and SVR. This could indicate that the underlying relationship between features and GPA in this dataset is largely linear, which explains the success of the simpler linear models. Random Forest, while typically robust for handling complex, non-linear interactions, performed worse than expected here, with both its R^2 value and MSE significantly behind the top models. This suggests that Random Forest may not be well-suited for this specific problem, or it may not have captured the subtle patterns that contribute to GPA as effectively as the other models.

AdaBoost and K-Nearest Neighbors (KNN), on the other hand, have the lowest R^2 values and the highest MSE and MAE, indicating that these models struggle to predict GPA accurately. KNN may suffer from issues related to scaling and distance-based learning in high-dimensional spaces, which could explain its poor performance.

D. Comparison with Previous Works:

The findings in this table are consistent with earlier studies that emphasize the strength of regularized linear models for predicting academic performance. For instance, several research papers highlight the importance of Ridge Regression and Lasso Regression for reducing overfitting and handling multicollinearity, which often improves prediction accuracy in educational datasets. XGBoost and Random Forest, although typically strong performers in many machine learning tasks, do not always outperform simpler models in structured data settings like this, where the relationship between the features and the target is relatively linear. Other studies have shown that, while XGBoost is powerful in complex datasets, linear models often provide competitive or superior performance when the dataset does not have highly non-linear patterns.

E. Verdict:

Based on the experimental results, Ridge Regression, Linear Regression, and SVR are the most suitable models for predicting GPA in this dataset. Their high R^2 values, combined with low MSE and MAE scores, indicate that they provide the most accurate and reliable predictions. Ensemble methods like XGBoost and Gradient Boosting offer slightly lower accuracy, suggesting that the underlying relationship in this dataset is predominantly linear. AdaBoost and KNN, on the other hand, show poor performance, making them unsuitable for this task. Overall, Ridge Regression emerges as the best model for this analysis due to its ability to balance model complexity and predictive accuracy.

V. Discussion

The study evaluated the performance of multiple regression models in predicting student GPA using R^2 scores, Mean Squared Error (MSE), and Mean Absolute Error (MAE). R^2 measures the model's ability to explain data variance, while MSE and MAE quantify prediction errors, with MAE reflecting the average absolute difference between predicted and actual values.

Ridge Regression and Linear Regression were the top performers, both achieving R^2 scores of 95.68%. This indicates their strong ability to explain data variance. The similarity in performance suggests that regularization in Ridge Regression doesn't significantly affect its accuracy. Both models also had the lowest MSE of 0.035696 and almost identical MAE values of 0.150637 and 0.150639, respectively, showcasing their high prediction accuracy and minimal deviation between predicted and actual GPA values.

Support Vector Regression (SVR) and XGBoost followed closely, with R^2 scores of 95.66% and 95.58%, respectively. These models handle nonlinear relationships effectively, explaining their strong results. Their MAE values were slightly higher, with SVR at 0.151026 and XGBoost at 0.145224, indicating a slightly higher error in individual predictions but still within an acceptable range.

Gradient Boosting, Lasso, and Random Forest also performed well, with R^2 scores ranging between 95.47% and 93.78%. These models, though slightly behind Ridge and Linear Regression, still showed strong predictive power. The MAE values for these models were 0.146663 (Gradient Boosting), 0.151654 (Lasso), and 0.167903 (Random Forest), indicating they made slightly larger average prediction errors compared to the top models.

At the lower end, AdaBoost and K-Nearest Neighbors (KNN) had R^2 scores of 92.63% and 89.02%, respectively, and showed a notable decline in performance. AdaBoost's MSE was 0.060884 with an MAE of 0.192798, while KNN had the highest MSE of 0.090838 and the largest MAE at 0.233688, indicating poor prediction accuracy and larger average errors for this dataset.

Overall, linear models like Ridge and Linear Regression performed the best, highlighting the linear relationship between the features and GPA. SVR and XGBoost showed strong results, but their added complexity may not be necessary given the success of simpler models. AdaBoost and KNN struggled due to their limitations in capturing underlying data relationships. In conclusion, Ridge Regression emerged as the best model, combining simplicity and accuracy for GPA prediction, with the lowest prediction errors (as evidenced by its low MSE and MAE values).

VI. Conclusion

The study compared several machine learning models for predicting student GPA, with the goal of identifying the most accurate and efficient model. Both Ridge Regression and Linear Regression achieved the highest $\square^2$ score of 95.68%, showing strong predictive accuracy. Support Vector Regression (SVR) and XGBoost also performed well, achieving $\square^2$ scores slightly lower than the linear models. Despite their complexity, these models did not substantially outperform Ridge or Linear Regression, which suggests that the relationship between features and GPA is mostly linear.

AdaBoost and K-Nearest Neighbors (KNN) were the weakest performers, with lower $\square^2$ scores and higher prediction errors, indicating their unsuitability for this dataset. The study concluded that Ridge Regression is the most effective model due to its simplicity and high accuracy, while more complex models like SVR and XGBoost, though capable of handling non-linear relationships, do not offer enough improvement to justify their complexity.

Thus, Ridge Regression is recommended for GPA prediction, providing a balance between performance, interpretability, and simplicity, with Linear Regression serving as a strong alternative. Models such as KNN and AdaBoost should be avoided in this context due to their poor performance.

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